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PAPER_1002: AGN Accretion Buoyancy-Corrected Eddington Luminosity

Abstract

The standard Eddington luminosity $L_{\text{Edd}} = 4\pi GMm_p c/\sigma_T$ is modified by SCm buoyancy. The correction factor $(1 + F_{\{\bar{U}, \text{Bi}\}}/(GM/r^2))$ accounts for vacuum buoyancy opposing gravitational collapse, raising the effective Eddington limit.

1. Corrected Eddington Luminosity

$$L_{\text{Edd}}^{\{\text{UQFF}\}} = L_{\text{Edd}} \cdot (1 + \rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)^2} / (GM/r_H^2))$$

For $M = 10^8 M_\odot$: $L_{\text{Edd}} = 1.26 \times 10^{39} \text{ W}$.

2. Implications

Super-Eddington accretion rates become permissible when buoyancy corrections reduce the effective radiation pressure barrier.

3. Implementation

File: fubi_agn_ns_mergers.py, class AGNAccretionBuoyancyCalc. CP4 class #586.